

Gibbs posterior distributions¹

construction, concentration, and calibration

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The idea that statistical problems do not have to be solved as one coherent whole is anathema to Bayesians but is liberating for frequentists –Larry Wasserman

- Quantities of interest often exist in the real world, not just as parameters in a posited statistical model
- e.g., quantiles, moments, optimal classifiers, ...
- One can posit a model and express the quantity of interest in terms of model parameters, but leads to *indirect* inference:
 - maybe lots of nuisance parameters
 - potential issues with marginalization²
 - risk of model misspecification bias

²Fraser (STS 2011), M. arXiv:1607.05051

- For non-Bayesians, direct inference is easy (Wasserman)
- What about a *direct* Bayesian-like approach?
- *Gibbs posterior distributions*
 - quantity of interest defined by minimizing risk
 - work with empirical risk instead of likelihood
- High-level features:
 - no nuisance parameters
 - no risk of model misspecification bias
 - good concentration properties
- Valid uncertainty quantification? No free lunch...

- Construction of Gibbs posteriors
- Asymptotic concentration properties
- Need for calibration and how to do it
- Examples will be interspersed throughout:
 - minimum clinically important difference (MCID)
 - quantile regression
 - classification
- Concluding remarks, open problems, etc.

- Suppose we have iid data $Z^n = (Z_1, \dots, Z_n)$ from P
- Z_i 's could be scalars, vectors, X - Y pairs, etc.
- Loss function $(z, \theta) \mapsto \ell_\theta(z)$ measures compatibility of (z, θ)
- Risk function, $R(\theta) = \int \ell_\theta(z) P(dz)$, expected loss
- Quantity of interest: $\theta^\star = \arg \min_\theta R(\theta)$
- Example:
 - $\ell_\theta(z) = |z - \theta|$
 - $\arg \min R(\theta) = \text{median of } P$
- Empirical risk: $R_n(\theta) = n^{-1} \sum_{i=1}^n \ell_\theta(Z_i)$

- Gibbs is Bayes but with a scaled empirical risk

$$\Pi_n^{(\eta)}(d\theta) \propto e^{-\eta n R_n(\theta)} \Pi(d\theta)$$

- Derivation in, e.g., Bissiri et al (JRSS-B, 2016)
- Direct attack on quantity of interest:
 - no nuisance parameters to set priors for
 - no marginalization
 - no statistical model, can't be misspecified
- Choice of learning rate η is crucial — data-driven?

Minimum clinically important difference

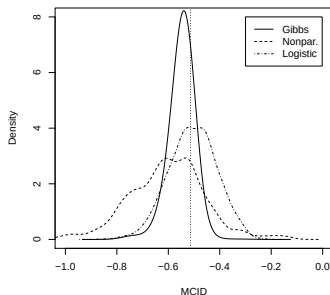
- Data $Z_i = (X_i, Y_i)$, $i = 1, \dots, n$, iid from P :³
 - $X_i \in \mathbb{R}$ is a diagnostic measure on patient i
 - $Y_i \in \{\pm 1\}$, depending on whether patient i found the treatment to be effective/ineffective
- Quantity of interest is the MCID:

$$\theta^* = \arg \min_{\theta} \underbrace{P\{Y \neq \text{sign}(X - \theta)\}}_{R(\theta)}$$

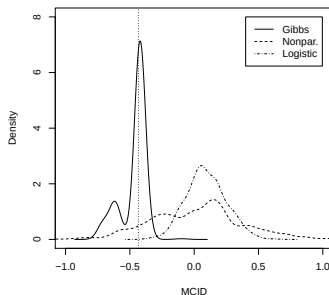
- MCID is the X -scale cutoff at which patients will tend to view the treatment as effective
- Statistical versus practical significance...

³Syring and M., arXiv:1501.01840

- Gibbs: $R_n(\theta) = n^{-1} \sum_{i=1}^n 1\{Y_i \neq \text{sign}(X_i - \theta)\}$.
- Compare Gibbs ($\eta = 1$) to two Bayes posteriors
 - logistic regression
 - nonparametric version



(a) "Good P case"



(b) "Bad P case"

- We expect that $\Pi_n^{(\eta)}$ will concentrate around θ^*
- Intuition is clear: $\Pi_n^{(\eta)}$ follows R_n and R_n follows R
- Formal statement and proof requires more care⁴
- I'll focus here on finite-dim cases — it's easier, more intuitive, and allows for static choices of η
- Other different approaches:
 - Grünwald and Mehta (JMLR 2020)
 - Syring and M., arXiv:2012.04505

⁴M. and Syring, arXiv:2203.09381

Proposition 1 – Rates

If there exists $\alpha > \beta > 0$ such that

$$\inf_{\theta: d(\theta, \theta^*) > \delta} \{R(\theta) - R(\theta^*)\} \gtrsim \delta^\alpha \quad \text{and} \quad \mathbb{E} \sup_{\theta: d(\theta, \theta^*) < \delta} |\mathbb{G}_n(\ell_\theta - \ell_{\theta^*})| \lesssim \delta^\beta$$

and if the prior “puts enough mass near θ^* ,” then

$$\Pi_n^{(\eta)}(\{\theta : d(\theta, \theta^*) \gtrsim n^{-1/(2\alpha-2\beta)}\}) \rightarrow 0, \quad \text{in P-probability}$$

Proposition 2 – BvM

Roughly, if the risk function is sufficiently smooth, etc., then

$$\Pi_n^{(\eta)} \approx \mathcal{N}(\hat{\theta}_n, \{\eta n \ddot{R}(\theta^*)\}^{-1}), \quad \text{for large } n$$

(M. and Syring, arXiv:2203.09381)

- Cases with $n^{-1/2}$ rates and a BvM:
 - mean regression
 - (multivariate⁵) quantiles
 - quantile regression
 - some classification problems
- MCID problem:⁶
 - Key quantity: $f(x) = P(Y = 1 \mid X = x)$
 - Roughly, if f is γ -Holder near θ^* , then rate $n^{-1/(1+2\gamma)}$
 - Specific cases:
 - if f is differentiable, then $\gamma = 1$ and rate is $n^{-1/3}$
 - if f has a jump at θ^* , then $\gamma = 0$ and rate is n^{-1} .
- High- or infinite-dim cases (e.g., *personalized* MCID⁷) too

⁵Bhattacharya and M., arXiv:2002.01052

⁶Syring and M., arXiv:1501.01840

⁷Syring and M., arXiv:2012.04505

- Asymptotic properties are nice, but there's a catch
- Unlike the *well-specified* Bayesian case, the above BvM does not imply good frequentist properties
- With a misspecified model, the Bayesian's results can be lousy⁸ — and there's nothing he can do about it
- But Gibbsians have options, through tuning η
- Not perfect/one-size-fits-all, but better than nothing!

⁸Kleijn and van der Vaart (EJS 2012)

- Recall the Gibbs posterior:

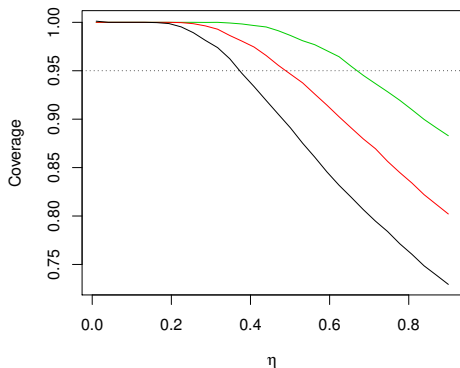
$$\Pi_n^{(\eta)}(d\theta) \propto e^{-\eta n R_n(\theta)} \Pi(d\theta).$$

- $\Pi_n^{(\eta)} \rightarrow \theta^*$ for “all” η , so η must control the spread
- η too big makes credible sets too small and vice versa
- Goal: valid Gibbs posterior uncertainty quantification
- Can we tune η so that $\Pi_n^{(\eta)}$ credible regions are calibrated, e.g., 95% credible regions are 95% confidence regions?⁹¹⁰

⁹Syring and M., arXiv:1509.00922

¹⁰Comparison to other methods in Wu and M., arXiv:2012.11349

MCID example: Coverage prob of 95% Gibbs posterior credible interval as a function of η for three n values

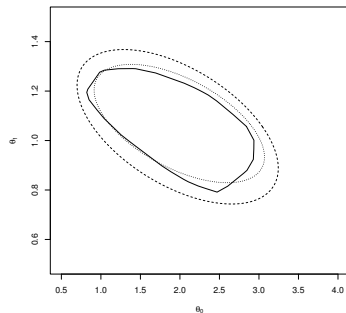


General Posterior Calibration (GPC) Algorithm:

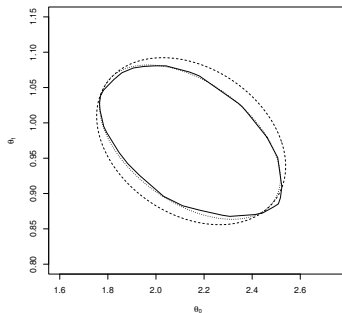
- 0 Initialize η_0 and set $t = 0$
- 1 Sample new data sets from $\mathbb{P}_n$ to approximate the coverage prob of Gibbs credible region $C_{\eta_t, \alpha}$:
 - do Gibbs posterior computation for each new data set
 - check coverage using $\hat{\theta}_n = \arg \min R_n(\theta)$
- 2 If empirical coverage prob is close to $1 - \alpha$, then stop; otherwise, update $\eta_t \rightarrow \eta_{t+1}$ via stochastic approximation, increment t , and go back to Step 1.

(Syring and M., arXiv:1509.00922)

Credible region contours: Quantile regression with two-dim θ

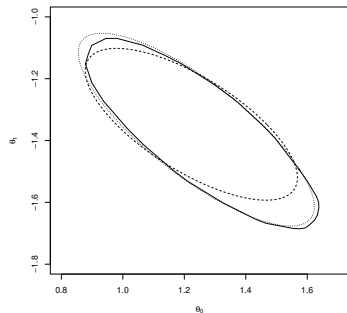


(c) $n = 50$, $\hat{\eta} \approx 0.96$

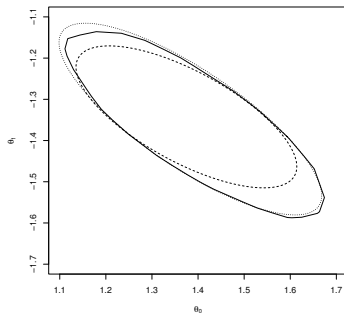


(d) $n = 400$, $\hat{\eta} \approx 0.90$

Credible region contours: Classification with two-dim θ



(e) $n = 400$, $\hat{\eta} \approx 0.70$



(f) $n = 1000$, $\hat{\eta} \approx 0.58$

- Two-dim quantile regression¹¹

BEL.s Bayesian empirical likelihood (Yang and He, AoS 2012)

Norm asymptotic normality

GPC Gibbs with GPC

- GPC gives exact coverage and shorter length!

n		Coverage Prob $\times 100$			Avg Length $\times 100$		
		BEL.s	Norm	GPC	BEL.s	Norm	GPC
100	θ_0	97	95	95	106	100	91
	θ_1	98	98	95	58	55	47
400	θ_0	95	95	95	50	50	46
	θ_1	97	97	95	26	25	23
1600	θ_0	96	96	95	25	25	23
	θ_1	96	96	95	13	12	11

¹¹Syring and M., arXiv:1509.00922

- Gibbs is direct/liberated Bayes:
 - no misspecification bias
 - no nuisance parameters
 - no marginalization
- Good concentration rate properties
- Can tune η for valid uncertainty quantification¹²
- Also possible to reverse engineer a suitable R :
 - image segmentation¹³
 - Lévy density estimation¹⁴
- More Gibbs stuff to do:
 - proof that the GPC algorithm “works”
 - low-dim structure learning
 - computation

¹²Also for prediction (Wu and M., arXiv:2107.01688)

¹³Syring and M., arXiv:1606.08400

¹⁴Wang and M., arXiv:2109.06567

Thanks for your attention!

- Questions? rgmarti3@ncsu.edu
 - Links to papers: www4.stat.ncsu.edu/~rmartin/
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